Benchmarking Medical VLMs for Radiology Image Captioning

Presented by :
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Github: https://github.com/krasniukevych-marichka/Benchmarking-Medical-Vision-Language-Models-for-Radiology-Image-Captioning/tree/main/results

Introduction

Problem and motivation: Growing volumes of CT, MRI and X-ray studies overload radiologists, causing burnout and delays in diagnostic reports. We benchmark medical Vision-Language Models that draft a radiology image description for the physician to review and approve. The AI acts as an assistant, not an autonomous diagnostician, cutting time spent on routine reporting.

Data:

ROCOv2 (Radiology Objects in COntext):

- **Images**: 79,789 radiology images (train 59,958 / val 9,904 / test 9,927)
- **Captions**: one text description per image, used as the reference for generation
- **Concepts**: multi-label UMLS CUI tags per image
  - 18 curated: 6 modalities (CT, X-ray, MRI, US, Angiogram, PET), 8 anatomical regions, 4 projections
  - full vocabulary: 1,947 CUIs, including findings



CUI concepts: C0024485; C0228465
Curated CUI concepts: C0024485
Caption: Enhanced magnetic resonance imaging of head revealed bilateral cerebral and cerebellar hemispheres abnormal meningeal enhancement.

Work breakdown:

- Mariia Krasniukevych - generating baseline captions of 4 VLMs
- Viktoriia Kukurudza - concept detector

Generating baseline captions

Models:

- 4 pretrained medical VLMs, used zero-shot
- Different model families
- Same prompt for all models

Issues

- 2,000 test images instead of 9,927: full run took too long, so we used a stratified subset
- LLaVA-Med in 4-bit: loaded quantized to fit GPU memory, so its scores may be slightly lower

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	CIDEr	ROUGE-L	Bertscore F1
Medmo	0.223	0.1404	0.0907	0.0599	0.605	0.2863	0.8851
MedGemma	0.2224	0.1179	0.0622	0.0324	0.1728	0.1906	0.8623
Hulu-med	0.1773	0.0924	0.0496	0.0266	0.0766	0.1623	0.8568
LlaVa-med	0.1214	0.0607	0.0294	0.0143	0.0203	0.134	0.8483

Concept detector

Model	micro-F1	macro-F1	sample-F1
ResNet-50 BCEWithLogitsLoss all concepts	0.548886	0.008650	0.577216
ResNet-50 BCEWithLogitsLoss 18 concepts	0.918345	0.797557	0.920586
ResNet-50 WeightedLoss all concepts	0.273663	0.040386	0.283817
ResNet-50 WeightedLoss 18 concepts	0.7100	0.5868	0.7408
ResNet-50 AssymetricLoss all concepts	0.3812	0.0385	0.3988
ResNet-50 AssymetricLoss 18 concepts	0.761930	0.637984	0.784814

Image Augmentation: Applied random crop, brightness/contrast adjustments and horizontal flips to handle visual variations and intensity differences across radiology scans.

- **Baseline (BCE):** Performs well on frequent classes (18-concept subset), but misses rare pathologies on full dataset.
- **Weighted Loss:** Increases rare class detection (higher Macro-F1), but severely suffers from high false positives.
- **Asymmetric Loss (ASL):** Best overall balance - significantly reduces false positives on full concepts and outperforms Weighted Loss across all metrics on the 18-concept subset.

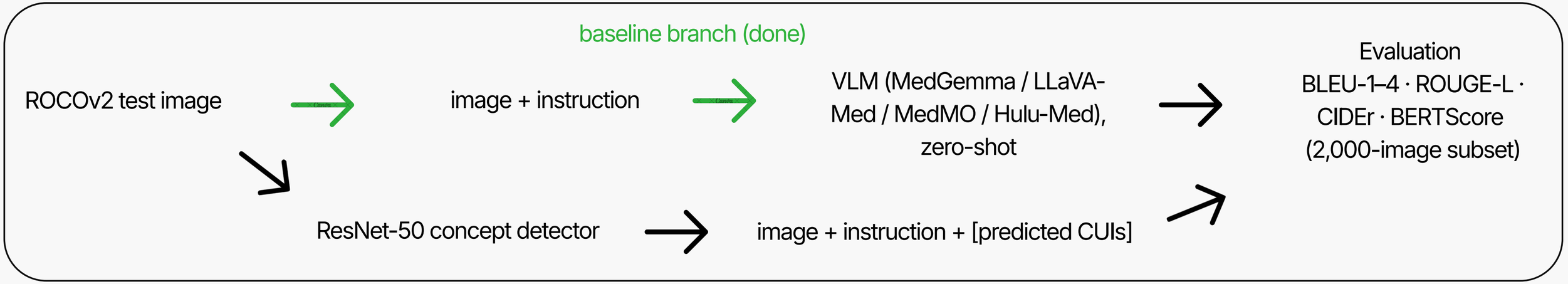
Before midterm:

- data preparation and EDA of ROCOv2 ✓
- training the baseline ResNet-50 concept detector on the full vocabulary (1,947 CUIs) ✓
- generating baseline captions from the four VLMs with the evaluation pipeline (BLEU, ROUGE-L, CIDEr, BERTScore) ✓

Next steps

- **Concept detector:** retrain ResNet-50 on the 18 curated CUIs only
- **Prompt integration:** add predicted concepts to the VLM prompt as context
- **Comparison:** 4 models × 2 settings (no concepts / 18 CUIs) on the same 2,000 images
- **Analysis:** do concepts improve captions? Which models benefit most? Conclusions and error analysis

Final pipeline



Risk

If incorporating predicted concepts into the prompt does not show a clear performance gain or leads to noisy descriptions, we will focus on experimenting with different prompt structures to find the most effective way to guide the models

UI Prototype

If successful, we will build an interactive web interface where users can upload radiological images, retrieve predicted concepts and generate draft descriptions that doctors can easily review and edit for practical clinical use.

Thank You

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